MT-1: Literature Extraction & Convention Fixing for Black Hole Entropy and Four Laws of Thermodynamics

From the SCT Spectral Action to Wald Entropy, $c_{\log}$, and Modified Hawking Radiation

David Alfyorov

March 2026 — Working Document

Credits: Aliaksandr Samatyia contributed research-assistance and workflow support.

Abstract

This document collects all formulas, conventions, and benchmarks from the literature needed to derive the black hole entropy from the SCT spectral action, compute the logarithmic correction $c_{\log}$, verify all four laws of black hole thermodynamics, and derive the modified Hawking radiation spectrum. The approach proceeds via three routes: (a) the Wald entropy formula applied to the nonlocal spectral action (using the NT-4b variational derivative $\delta S_{\text{spec}}/\delta R_{\mu\nu\rho\sigma}$); (b) the Sen formula for logarithmic corrections using the SM field content established in NT-1b Phase 3; (c) the modified Regge–Wheeler/Zerilli equations for greybody factors. All formulas are extracted from the literature with source annotations; no original derivations are performed. Gaps requiring resolution in the derivation phase are noted explicitly.

Key references:

- Wald (1993): Phys. Rev. D **48**, R3427. BH entropy as Noether charge.
- Iyer–Wald (1994): Phys. Rev. D **50**, 846. Properties of Noether charge and generalization to arbitrary diffeomorphism-invariant theories.
- Jacobson–Myers (1993): Phys. Rev. Lett. **70**, 3684. Entropy of Lovelock and higher-curvature gravity.
- Sen (2012): arXiv:1205.0971. Logarithmic corrections to Schwarzschild and other non-extremal BH entropy.
- Wall (2015): Int. J. Mod. Phys. D **24**, 1544014; also arXiv:1504.08040. Second law for higher-derivative gravity.
- Mann–Solodukhin (1998): Nucl. Phys. B **523**, 293; arXiv:hep-th/9709064. Conical singularity method.
- Frolov–Fursaev (1997): Phys. Rev. D **56**, 2212. Thermal fields and BH entropy.
- Buoninfante–Mazumdar (2020): arXiv:1903.01542; also Buoninfante–Modesto–Shapiro 2001.07830. Nonlocal gravity and BH thermodynamics.

- Connes–Chamseddine (2010): arXiv:1004.0464. Spectral action and gravity.
- Stelle (1977): Phys. Rev. D **16**, 953; Gen. Rel. Grav. **9** (1978) 353.
- Calcagni–Modesto (2022): arXiv:2211.05606. Nonlocal field equations.

Contents

1 Background: From NT-4b to MT-1 **4**
- 1.1 What NT-4b established . . . 4
- 1.2 What MT-1 must achieve . . . 4
- 1.3 Conventions inherited from NT-1 through NT-4b . . . 4

2 Schwarzschild Geometry **5**
- 2.1 Metric and curvature . . . 5
- 2.2 Horizon properties . . . 5
- 2.3 Bifurcation surface and binormal . . . 5

3 The Wald Entropy Formula **6**
- 3.1 General statement . . . 6
- 3.2 Einstein–Hilbert contribution . . . 6
- 3.3 Weyl-squared (C^2) contribution . . . 7
- 3.4 R^2 contribution . . . 7
- 3.5 Nonlocal corrections . . . 8

4 Full Wald Entropy in SCT **8**

5 Logarithmic Correction $c_{\log}$ **8**
- 5.1 Sen's formula . . . 8
- 5.2 SCT prediction for $c_{\log}$ . . . 9
- 5.3 Relation to conformal anomaly . . . 10

6 Four Laws of BH Thermodynamics in SCT **10**
- 6.1 Zeroth law: constancy of surface gravity . . . 10
- 6.2 First law: energy balance . . . 11
- 6.3 Second law: area non-decrease . . . 11
- 6.4 Third law: extremal limit . . . 12

7 Modified Hawking Radiation **12**
- 7.1 Standard Hawking spectrum . . . 12
- 7.2 Greybody factors and the Regge–Wheeler equation . . . 12
- 7.3 SCT modifications to the effective potential . . . 13

8 Entanglement Entropy Approach **13**

9 Benchmarks and Cross-Checks **14**

10 Gaps and Open Questions **14**

11 Master Reference Table **15**

12 Notation Summary **16**

13 Summary: The Full BH Entropy Formula in SCT **16**

1 Background: From NT-4b to MT-1

1.1 What NT-4b established

NT-4b derived the full nonlinear field equations of the SCT spectral action on arbitrary curved backgrounds. The key results relevant for MT-1 are:

- **Full field equations:**

$$\frac{1}{\kappa^2}\,G_{\mu\nu}+\frac{1}{16\pi^2}\Big[2\,F_1\Big(\frac{\Box}{\Lambda^2}\Big)\,B_{\mu\nu}+\Theta^{(C)}_{\mu\nu}\Big]+\frac{1}{16\pi^2}\Big[F_2\Big(\frac{\Box}{\Lambda^2},\xi\Big)\,H_{\mu\nu}+\Theta^{(R)}_{\mu\nu}\Big]=\tfrac{1}{2}\,T_{\mu\nu}. \quad (1)$$

- **Ricci-flat simplification:** On Schwarzschild/Kerr, $R=0$, $R_{\mu\nu}=0$, so $H_{\mu\nu}=0$, $\Theta^{(R)}_{\mu\nu}=0$, and only the Weyl sector contributes.
- **Wald variation:**

$$\frac{\partial\mathcal{L}_{\text{quad}}}{\partial R_{\mu\nu\rho\sigma}}=\frac{1}{16\pi^2}\Big[2F_1(0)\,\frac{\partial C^2}{\partial R_{\mu\nu\rho\sigma}}+F_2(0,\xi)\,\frac{\partial R^2}{\partial R_{\mu\nu\rho\sigma}}\Big], \quad (2)$$

  where on the horizon $F_i(\Box/\Lambda^2)\to F_i(0)$ for astrophysical BH ($r_H\gg 1/\Lambda$).
- **Correction scale:** $\mathcal{O}(\Lambda^2/M_{\text{Pl}}^2)\sim\mathcal{O}(10^{-60})$ for $\Lambda\sim$ eV.

1.2 What MT-1 must achieve

From the roadmap (Section MT-1):

1. Derive $S_{\text{BH}}=A/(4G)$ from the spectral action via the Wald entropy formula.
2. Compute the logarithmic correction $c_{\log}$ as a parameter-free prediction.
3. Verify all four laws of BH thermodynamics within SCT.
4. Derive the modified Hawking radiation spectrum (greybody factors).

1.3 Conventions inherited from NT-1 through NT-4b

Symbol	Definition
$g_{\mu\nu}$	Lorentzian metric, signature $(-,+,+,+)$
$R^{\rho}{}_{\sigma\mu\nu}$	Riemann tensor
$C_{\mu\nu\rho\sigma}$	Weyl tensor
κ^2	$8\pi G=8\pi/M_{\text{Pl}}^2$
Λ	Spectral cutoff scale
ξ	Higgs non-minimal coupling to gravity
α_C	$13/120$ (total Weyl^2 coefficient, parameter-free)
$\alpha_R(\xi)$	$2(\xi-1/6)^2$ (total R^2 coefficient)
$F_1(0)$	$\alpha_C/(16\pi^2)=13/(1920\pi^2)$
$F_2(0,\xi)$	$\alpha_R(\xi)/(16\pi^2)$
m_2	$\Lambda\sqrt{60/13}$ (spin-2 effective mass)
$m_0(\xi)$	$\Lambda/\sqrt{6(\xi-1/6)^2}$ (spin-0 effective mass)
$\varphi(x)$	$e^{-x/4}\sqrt{\pi/x}\,\text{erfi}(\sqrt{x}/2)$ (master function, entire)

Convention 1.1 (SM field content). [CPR 0805.2909; NT-1b Phase 3.]

Field	Spin	Count	Convention
Real scalars (Higgs)	0	$N_s = 4$	Real components
Weyl fermions	1/2	$N_f = 45$	Weyl count
Dirac fermions	1/2	$N_D = N_f/2 = 22.5$	Dirac count
Gauge bosons	1	$N_v = 12$	$SU(3) \times SU(2) \times U(1)$

2 Schwarzschild Geometry

2.1 Metric and curvature

[MTW (1973); Wald (1984), Chapter 6.]

The Schwarzschild metric in standard coordinates:

$$\mathrm{d}s^2 = -f(r)\,\mathrm{d}t^2 + f(r)^{-1}\,\mathrm{d}r^2 + r^2\,\mathrm{d}\Omega^2, \qquad f(r) = 1 - \frac{r_H}{r}, \qquad r_H = 2GM. \tag{3}$$

Proposition 2.1 (Schwarzschild curvature invariants)**.** *[Wald (1984), eq. (6.1.7)–(6.1.11).]*

$$R = 0, \qquad R_{\mu\nu} = 0 \quad (\textit{vacuum}), \tag{4}$$

$$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = \frac{48\,G^2M^2}{r^6}, \tag{5}$$

$$C_{\mu\nu\rho\sigma} = R_{\mu\nu\rho\sigma} \quad (\textit{since Ricci-flat}), \tag{6}$$

$$C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}\Big|_{r=r_H} = \frac{48\,G^2M^2}{(2GM)^6} = \frac{3}{4\,G^4M^4}. \tag{7}$$

2.2 Horizon properties

[Wald (1984); Poisson (2004).]

Quantity	Value
Horizon radius	$r_H = 2GM$
Surface gravity	$\kappa = f'(r_H)/2 = 1/(4GM)$
Area	$A = 4\pi r_H^2 = 16\pi G^2M^2$
Hawking temperature	$T_H = \kappa/(2\pi) = 1/(8\pi GM)$
Bekenstein–Hawking entropy	$S_{\mathrm{BH}} = A/(4G) = 4\pi GM^2$

2.3 Bifurcation surface and binormal

[Wald (1993), eq. (12)–(14); Iyer–Wald (1994), eq. (8).]

The bifurcation 2-sphere $\mathcal{H}$ is the $(d-2)$-surface at which the Killing horizon of $\xi^a = (\partial/\partial t)^a$ has vanishing norm. The binormal $\hat{\epsilon}_{\mu\nu}$ is the antisymmetric tensor normal to $\mathcal{H}$, lying in the (t, r) plane:

$$\hat{\epsilon}_{\mu\nu} = 2\,n_{[\mu}\,s_{\nu]}, \qquad \hat{\epsilon}_{\mu\nu}\hat{\epsilon}^{\mu\nu} = -2, \tag{8}$$

where n^μ is the future-directed unit timelike normal and s^μ is the outward-directed unit spacelike normal to $\mathcal{H}$.

Proposition 2.2 (Weyl contraction on Schwarzschild bifurcation surface)**.** *[Jacobson–Myers (1993); Padmanabhan (2010), eq. (15.62).]*

In the orthonormal frame $(e^{\hat{0}}, e^{\hat{1}}, e^{\hat{2}}, e^{\hat{3}})$ *adapted to the* (t, r) *binormal:*

$$C_{\mu\nu\rho\sigma}\,\hat{\epsilon}^{\mu\nu}\hat{\epsilon}^{\rho\sigma}\Big|_{\mathcal{H}} = 4\,C_{\hat{0}\hat{1}\hat{0}\hat{1}}\Big|_{r=r_H} = 4\cdot\left(-\frac{2GM}{r_H^3}\right) = -\frac{1}{G^2M^2}. \tag{9}$$

Remark 2.3 (Uniformity on the 2-sphere). The Schwarzschild Weyl tensor has $\ell = 0$ angular dependence in the orthonormal frame (it is spherically symmetric), so the integrand in the Wald formula is constant over $\mathcal{H}$. The integral over S^2 simply multiplies by the area $A = 4\pi r_H^2$.

3 The Wald Entropy Formula

3.1 General statement

[Wald (1993), Phys. Rev. D **48**, R3427; Iyer–Wald (1994), Phys. Rev. D **50**, 846.]

For any diffeomorphism-invariant Lagrangian $L(g_{\mu\nu}, R_{\mu\nu\rho\sigma}, \nabla_\alpha R_{\mu\nu\rho\sigma}, \ldots)$, the black hole entropy is

$$\boxed{S_{\text{Wald}} = -2\pi \oint_{\mathcal{H}} \frac{\partial\mathcal{L}}{\partial R_{\mu\nu\rho\sigma}}\,\hat{\epsilon}_{\mu\nu}\,\hat{\epsilon}_{\rho\sigma}\,\mathrm{d}A.} \tag{10}$$

This formula gives the correct first law of black hole mechanics for *all* diffeomorphism-invariant theories.

Remark 3.1 (Covariant derivatives of Riemann). When the Lagrangian depends on $\nabla_\alpha R_{\mu\nu\rho\sigma}$ (as in our nonlocal spectral action), the Wald formula generalizes via integration by parts. For a Lagrangian of the form $\mathcal{L} = f(\Box)\,I$ where I is a curvature invariant and $\Box = g^{\mu\nu}\nabla_\mu\nabla_\nu$, the on-horizon evaluation uses the local limit $f(\Box) \to f(0)$ for astrophysical BH ($r_H \gg 1/\Lambda$), and additional ∇-dependent corrections are suppressed by powers of $\kappa^2/\Lambda^2 \ll 1$. [Buoninfante–Mazumdar (2020); Conroy–Edholm–Mazumdar (2017), arXiv:1509.01636.]

3.2 Einstein–Hilbert contribution

[Wald (1993), eq. (14); Iyer–Wald (1994), eq. (16).]

For $\mathcal{L}_{\text{EH}} = R/(16\pi G)$:

$$\frac{\partial\mathcal{L}_{\text{EH}}}{\partial R_{\mu\nu\rho\sigma}} = \frac{1}{16\pi G}\,\frac{\partial R}{\partial R_{\mu\nu\rho\sigma}} = \frac{1}{16\pi G}\,g^{\mu[\rho}\,g^{\sigma]\nu}. \tag{11}$$

Contracting with the binormal:

$$g^{\mu[\rho}\,g^{\sigma]\nu}\,\hat{\epsilon}_{\mu\nu}\,\hat{\epsilon}_{\rho\sigma} = -2, \tag{12}$$

so

$$S_{\text{EH}} = -2\pi\cdot\frac{1}{16\pi G}\cdot(-2)\cdot A = \frac{A}{4G}. \tag{13}$$

This is the standard Bekenstein–Hawking result.

3.3 Weyl-squared (C^2) contribution

[Jacobson–Myers (1993), eq. (7); Jacobson–Kang–Myers (1994), arXiv:gr-qc/9312023.]

For $\mathcal{L}_{C^2} = \alpha\, C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}$ where $\alpha = \alpha_C/(16\pi^2)$ [our convention]:

$$\frac{\partial \mathcal{L}_{C^2}}{\partial R_{\mu\nu\rho\sigma}} = \alpha\, \frac{\partial (C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma})}{\partial R_{\mu\nu\rho\sigma}}. \tag{14}$$

The Weyl tensor decomposition is

$$C_{\mu\nu\rho\sigma} = R_{\mu\nu\rho\sigma} - \Big(g_{\mu[\rho}\, R_{\sigma]\nu} - g_{\nu[\rho}\, R_{\sigma]\mu}\Big) + \tfrac{1}{3}\, R\, g_{\mu[\rho}\, g_{\sigma]\nu}. \tag{15}$$

On a Ricci-flat background ($R = 0$, $R_{\mu\nu} = 0$):

$$C_{\mu\nu\rho\sigma}\Big|_{\text{Ricci-flat}} = R_{\mu\nu\rho\sigma}, \qquad C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}\Big|_{\text{Ricci-flat}} = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}. \tag{16}$$

Proposition 3.2 (Jacobson–Myers formula on Ricci-flat backgrounds)**.** *[Jacobson–Myers (1993), eq. (7)–(10).]*

For $\mathcal{L} = \gamma\, R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ (Gauss-Bonnet-type term in the Riemann-squared sector), the Wald entropy correction is

$$\delta S_{\text{JM}} = 8\pi\gamma\, \chi(\mathcal{H}), \tag{17}$$

where $\chi(\mathcal{H})$ is the Euler characteristic of the horizon cross-section. For $\mathcal{H} = S^2$: $\chi = 2$.

Corollary 3.3 (Wald entropy correction from C^2 on Schwarzschild)**.** *On Schwarzschild, $C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ and the coefficient of $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ in the action is $\gamma = \alpha = \alpha_C/(16\pi^2)$. The Jacobson–Myers formula (17) gives:*

$$\boxed{\delta S_{C^2} = 8\pi \cdot \frac{\alpha_C}{16\pi^2} \cdot 2 = \frac{\alpha_C}{\pi} = \frac{13}{120\pi} \approx 0.0345.} \tag{18}$$

*This is a **topological** correction (independent of the horizon area A), arising from the Euler characteristic of S^2.*

Remark 3.4 (Area-independent correction). The C^2 Wald entropy correction is $\alpha_C/\pi \approx 0.035$, which is a dimensionless constant of order unity. For an astrophysical black hole with $S_{\text{BH}} \sim 10^{79}$, this correction is utterly negligible. The correction becomes relevant only for micro-black holes with $A \sim \ell_P^2$, where higher-order terms also become important.

3.4 R^2 contribution

[Jacobson–Myers (1993); Padmanabhan (2010), Chapter 15.]

For $\mathcal{L}_{R^2} = \beta\, R^2$ with $\beta = \alpha_R(\xi)/(16\pi^2)$:

$$\frac{\partial (R^2)}{\partial R_{\mu\nu\rho\sigma}} = 2R\, g^{\mu[\rho}\, g^{\sigma]\nu}. \tag{19}$$

Proposition 3.5 (R^2 contribution vanishes on Schwarzschild)**.** *On Schwarzschild, $R = 0$, so*

$$\delta S_{R^2}\Big|_{\text{Schw}} = 0. \tag{20}$$

The R^2 sector contributes to the Wald entropy only on backgrounds with nonzero Ricci scalar (de Sitter, Kerr–de Sitter, Reissner–Nordström–de Sitter, etc.).

3.5 Nonlocal corrections

[GAP]

The full SCT action contains nonlocal form factors $F_i(\Box/\Lambda^2)$, and the Wald formula for nonlocal actions requires additional care.

Proposition 3.6 (Local limit on astrophysical horizons)**.** *[Conroy–Edholm–Mazumdar (2017), arXiv:1509.01636, Section 5; Buoninfante–Mazumdar (2020), arXiv:1903.01542, Section 3.]*

For a Schwarzschild BH with mass $M \gg M_{\rm Pl}$:

- *The curvature scale on the horizon is $\kappa_H \sim 1/(GM) \ll \Lambda$.*
- *The d'Alembertian eigenvalue on the horizon satisfies $\Box/\Lambda^2 \sim \kappa_H^2/\Lambda^2 \ll 1$.*
- *The form factors reduce to their local ($z = 0$) values: $F_i(\Box/\Lambda^2) \to F_i(0)$.*
- *Nonlocal corrections to the Wald entropy are suppressed by $(\kappa_H/\Lambda)^2 \sim (M_{\rm Pl}/M)^2 \cdot (\Lambda/M_{\rm Pl})^2$.*

Remark 3.7 (Near-Planckian regime). For micro-BH with $M \sim M_{\rm Pl}$ or $r_H \sim 1/\Lambda$, the nonlocal structure of $F_1(z)$ becomes important and the local Wald formula does not apply directly. In this regime, the entire-function property of $F_1(z)$ (from NT-2) prevents pole-induced divergences, but the full nonlocal Wald formula must be used. [GAP] *The full nonlocal Wald formula for $\int F(\Box)\, C^2$ requires resolution in the derivation phase.*

4 Full Wald Entropy in SCT

Combining the Einstein–Hilbert contribution (§3), the C^2 correction (Corollary 3.3), and the vanishing R^2 contribution (Proposition 3.5):

Theorem 4.1 (Wald entropy of Schwarzschild BH in SCT)**.** *For a Schwarzschild BH in the SCT spectral action, the Wald entropy is*

$$\boxed{S_{\rm Wald} = \frac{A}{4G} + \frac{\alpha_C}{\pi} = \frac{A}{4G} + \frac{13}{120\pi}.} \tag{21}$$

The first term is the Bekenstein–Hawking area law. The second term is a topological correction from the Weyl-squared sector, determined entirely by the SM field content via $\alpha_C = 13/120$.

For Kerr–Newman BH with nonzero charge/angular momentum:

$$S_{\rm Wald} = \frac{A}{4G} + \frac{\alpha_C}{\pi}\,\chi(\mathcal{H}) + \frac{\alpha_R(\xi)}{\pi}\,\delta S_{R^2}(\xi), \tag{22}$$

where $\chi(\mathcal{H}) = 2$ for topologically spherical horizons and δS_{R^2} is nonzero only when $R \neq 0$ (e.g., in de Sitter backgrounds).

5 Logarithmic Correction $c_{\log}$

5.1 Sen's formula

[Sen (2012), arXiv:1205.0971, eq. (1.2) and Table 1.]

For a 4-dimensional Schwarzschild BH in Einstein gravity coupled to n_S real scalars, n_F massless **Dirac** fermions, and n_V massless vectors, the one-loop logarithmic correction to the microcanonical entropy is

$$\boxed{c_{\log} = \frac{1}{180}\Big[2\,n_S + 7\,n_F - 26\,n_V + 424\Big].} \tag{23}$$

The +424 accounts for the graviton contribution (including zero modes). The vector coefficient is −26 (negative, due to Faddeev–Popov ghost dominance in the Kretschner term of the Seeley–DeWitt a_4 coefficient).

[Sign correction] Corrected from LR audit. The original extraction had the vector and graviton signs reversed and used Weyl fermion counting with coefficient 7. The correct Sen formula uses **Dirac** fermions with coefficient 7, vector coefficient −26, and graviton +424. Verified against Sen (2012) eq. (1.2) and independently against arXiv:2104.14902 eq. (5).

Remark 5.1 (Derivation method). Sen's result is obtained via the Euclidean gravity method (heat kernel on the Euclidean Schwarzschild instanton), with careful treatment of the zero modes of the graviton kinetic operator. The computation uses the Seeley–DeWitt coefficients $a_4(x)$ evaluated on the instanton background, integrated against the logarithmic divergence $\ln(\Lambda^2 A)$. The individual field contributions are:

Field	Contribution to $180\,c_{\log}$
Real scalar	+2 per field
Dirac fermion	+7 per field
Massless vector	−26 per field
Graviton	+424 (total)

Remark 5.2 (Weyl fermion convention). The formula can be rewritten in terms of Weyl fermions ($n_W = 2\,n_F$):

$$c_{\log} = \frac{1}{180}\Big[2\,n_S + \tfrac{7}{2}\,n_W - 26\,n_V + 424\Big]. \tag{24}$$

This is equivalent to (23) by the substitution $n_F = n_W/2$. **Note:** the coefficient per Weyl fermion is 7/2, not 7.

5.2 SCT prediction for $c_{\log}$

Using the SM field content from Convention 1.1 ($n_S = 4$, $n_F = N_D = 45/2 = 22.5$ Dirac fermions, $n_V = 12$):

Theorem 5.3 (Parameter-free prediction of $c_{\log}$ in SCT)**.**

$$\boxed{c_{\log} = \frac{1}{180}\Big[2\cdot 4 + 7\cdot\tfrac{45}{2} - 26\cdot 12 + 424\Big] = \frac{37}{24} \approx 1.542.} \tag{25}$$

*This is the **microcanonical** coefficient (no zero-mode subtraction).*

Verification. $2\cdot 4 = 8$; $7\cdot 22.5 = 157.5$; $-26\cdot 12 = -312$; graviton: +424. Total $= 8 + 157.5 - 312 + 424 = 277.5$. $c_{\log} = 277.5/180 = 37/24$. □

Remark 5.4 (Matter vs. graviton decomposition). The graviton contributes $+424/180 = 106/45 \approx 2.356$ (positive: gravity *increases* the logarithmic correction). The matter contribution is $(8 + 157.5 - 312)/180 = -146.5/180 \approx -0.814$ (net negative because the 12 vectors dominate over scalars and fermions). The net result is positive: $c_{\log} = 37/24 > 0$, meaning quantum corrections *increase* the entropy relative to the area law.

Remark 5.5 (Comparison with competing approaches).

Approach	$c_{\log}$ (4D Schwarzschild)	Reference
SCT (this work)	$\mathbf{37/24 \approx 1.542}$	**parameter-free**
LQG (isolated horizons)	$-3/2 = -1.5$	Kaul–Majumdar (2000)
String theory	model-dependent	Sen (2012)
Asymptotic safety	not well-defined	(running G)
IDG (Modesto)	same as Sen formula	Modesto (2012)

[Sign correction] The LQG prediction $c_{\log} = -3/2$ has the *opposite sign* from the SCT prediction. This is a potentially discriminating observable if logarithmic corrections to BH entropy become accessible, though this is far beyond current experimental capability. Sen (2012) notes that the LQG result disagrees with the Euclidean gravity computation.

5.3 Relation to conformal anomaly

[Christensen–Duff (1978), Nucl. Phys. B **154**, 301.]

The coefficients in Sen's formula are related to the Seeley–DeWitt coefficient a_4, which in turn determines the conformal (trace) anomaly:

$$\langle T^{\mu}{}_{\mu} \rangle = \frac{1}{16\pi^2}\Big[c\, C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} - a\, E_4 + b\,\Box R\Big], \tag{26}$$

where $E_4 = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 4R_{\mu\nu}R^{\mu\nu} + R^2$ is the Euler density and a, c are the central charges. The relationship to our α_C is:

$$c = \frac{1}{120}\Big[N_s + 6N_D + 12N_v\Big], \qquad a = \frac{1}{360}\Big[N_s + \tfrac{11}{2}\cdot 2N_D + 62N_v\Big]. \tag{27}$$

[needs check] The exact mapping between $c_{\log}$ and (a, c) involves additional contributions from the graviton sector and zero-mode subtleties. The Sen formula (23) incorporates all these corrections; it does not simply reduce to $c_{\log} = f(a, c)$.

6 Four Laws of BH Thermodynamics in SCT

6.1 Zeroth law: constancy of surface gravity

[Bardeen–Carter–Hawking (1973), Commun. Math. Phys. **31**, 161; Wald (1984), Section 12.5.]

Theorem 6.1 (Zeroth law – standard GR)**.** *For a stationary black hole satisfying the dominant energy condition, the surface gravity κ is constant on the event horizon.*

Status in SCT: The zeroth law depends on the field equations through the dominant energy condition (DEC). The SCT field equations (1) modify the Einstein equations, but for Ricci-flat backgrounds (Schwarzschild, Kerr), the modifications are $\mathcal{O}(\Lambda^2/M_{\rm Pl}^2)$, and the zeroth law holds to the same accuracy.

[GAP] For non-vacuum BH (e.g., charged or hairy), verification requires checking the DEC for the effective stress-energy tensor including the higher-derivative contributions.

6.2 First law: energy balance

[Wald (1993); Iyer–Wald (1994), eq. (27)–(30).]

For the SCT action, the first law takes the form

$$\mathrm{d}M = \frac{\kappa_{\mathrm{eff}}}{2\pi}\,\mathrm{d}S_{\mathrm{Wald}} + \Omega_H\,\mathrm{d}J + \Phi_H\,\mathrm{d}Q, \tag{28}$$

where S_{Wald} is the full Wald entropy (21) and κ_{eff} is the effective surface gravity.

Proposition 6.2 (First law for Wald entropy)**.** *[Iyer–Wald (1994), Theorem 1.] For any diffeomorphism-invariant Lagrangian, the first law* $\mathrm{d}M = (\kappa/2\pi)\,\mathrm{d}S_{\mathrm{Wald}} + \cdots$ *holds* exactly*, provided one uses the Wald entropy (not the area). This is a consequence of the Noether charge construction and does not require the DEC.*

Remark 6.3 (Modified first law in NCG). A 2025 result (arXiv:2509.23423, referenced in the SCT roadmap) shows that the conventional first law is violated in noncommutative geometry because the entropy deviates from Bekenstein–Hawking. In SCT, the first law is *maintained* precisely because we use the Wald entropy, which is the Noether charge. The "violation" in NCG arises from using the area entropy $A/(4G)$ instead of the Wald entropy.

6.3 Second law: area non-decrease

[Wall (2015), arXiv:1504.08040; Wall (2024), arXiv:2312.17636.]

Theorem 6.4 (Wall's second law for higher-derivative gravity)**.** *[Wall (2015), Theorem 1; Wall (2024).] For a diffeomorphism-invariant theory of gravity in $d \geq 4$ dimensions, the generalized second law*

$$\mathrm{d}S_{\mathrm{gen}} = \mathrm{d}S_{\mathrm{Wald}} + \mathrm{d}S_{\mathrm{outside}} \geq 0 \tag{29}$$

holds provided:

1. *The theory has a well-posed initial value problem.*
2. *The null energy condition holds for the matter sector.*
3. *The entropy functional satisfies certain linearized stability conditions.*

Status in SCT:

1. **Well-posedness:** The SCT field equations include higher-derivative terms, which generically threaten well-posedness. However, the entire-function property of $F_1(z)$ (NT-2, ghost-freedom) and the MR-3 causality analysis suggest that the initial value problem is well-posed at least for macroscopic perturbations. [GAP] Full proof of well-posedness for the nonlocal SCT equations is an open problem (related to MR-4).
2. **Null energy condition:** The SM matter sector satisfies the NEC classically.
3. **Linearized stability:** The spin-2 propagator denominator $\Pi_{\mathrm{TT}}(z)$ has ghost poles (MR-2), which may violate the stability condition. [GAP] This requires careful analysis in the derivation phase.

[**Sign correction**] The second law is CONDITIONAL on the ghost resolution (MR-2). If the ghosts are resolved via the Lee–Wick/fakeon prescription (FK analysis), the effective theory may satisfy Wall's conditions in the macroscopic regime where ghost effects are suppressed by $(M_{\mathrm{Pl}}/\Lambda)^2$.

6.4 Third law: extremal limit

[Israel (1986), Phys. Rev. Lett. **57**, 397; Wald (1997), arXiv:gr-qc/9702022.]

Theorem 6.5 (Third law – Nernst version)**.** *It is impossible to reduce the surface gravity κ of a black hole to zero by any finite physical process.*

Status in SCT: For the Schwarzschild BH, $\kappa = 1/(4GM)$ vanishes only as $M \to \infty$, which is not an extremal limit. For Kerr–Newman BH, the extremal condition $\kappa = 0$ occurs at $M^2 = a^2 + Q^2$ (where $a = J/M$). The SCT corrections to the extremal condition are:

$$M_{\text{ext}}^2 = a^2 + Q^2 + \mathcal{O}\left(\frac{\Lambda^2}{M_{\text{Pl}}^2}\right) \cdot (a^2 + Q^2). \tag{30}$$

The correction is negligible for astrophysical parameters.

[GAP] The Wald entropy at extremality and the attractor mechanism in the SCT context require further analysis (connects to MR-9).

7 Modified Hawking Radiation

7.1 Standard Hawking spectrum

[Hawking (1975), Commun. Math. Phys. **43**, 199; Page (1976), Phys. Rev. D **13**, 198.]

The Hawking temperature of a Schwarzschild BH is

$$T_H = \frac{\kappa}{2\pi} = \frac{1}{8\pi GM}. \tag{31}$$

The emission rate per unit frequency for a species of spin s is

$$\frac{\mathrm{d}N^{(s)}}{\mathrm{d}\omega\,\mathrm{d}t} = \frac{1}{2\pi}\,\frac{\Gamma_\ell^{(s)}(\omega)}{e^{\omega/T_H} \pm 1}, \tag{32}$$

where $\Gamma_\ell^{(s)}(\omega)$ is the greybody factor (absorption probability) for partial wave ℓ and spin s, and $+$ applies to fermions, $-$ to bosons.

7.2 Greybody factors and the Regge–Wheeler equation

[Regge–Wheeler (1957), Phys. Rev. **108**, 1063; Zerilli (1970), Phys. Rev. D **2**, 2141.]

In the standard Regge–Wheeler formalism, the radial wave equation for spin-s perturbations of Schwarzschild is

$$\frac{\mathrm{d}^2\Psi}{\mathrm{d}r_*^2} + \left[\omega^2 - V_{\text{eff}}^{(s)}(r)\right]\Psi = 0, \tag{33}$$

where $r_* = r + r_H \ln(r/r_H - 1)$ is the tortoise coordinate and

$$V_{\text{eff}}^{(s)}(r) = f(r)\left[\frac{\ell(\ell+1)}{r^2} + \frac{(1-s^2)\,r_H}{r^3}\right]. \tag{34}$$

7.3 SCT modifications to the effective potential

[GAP]

In SCT, the linearized equations around the Schwarzschild background are modified by the nonlocal form factors. The spin-2 perturbation equation becomes (schematically):

$$\Pi_{\mathrm{TT}}\left(\frac{-\partial_t^2+\nabla_r^2}{\Lambda^2}\right)\left[\frac{\mathrm{d}^2\Psi}{\mathrm{d}r_*^2}+\omega^2\,\Psi\right]-V_{\mathrm{eff}}^{(2)}(r)\,\Psi=0. \tag{35}$$

Proposition 7.1 (SCT modifications to greybody factors)**.** *[Conroy–Koshelev–Mazumdar (2017), arXiv:1605.02080, Section 4; Buoninfante et al. (2020), arXiv:1906.11238, Section 3.]*

- *At $\omega \ll \Lambda$: $\Pi_{\mathrm{TT}} \to 1$ and the standard Regge–Wheeler equation is recovered. Modifications to greybody factors are exponentially suppressed as $\sim e^{-\Lambda r_H}$.*
- *At $\omega \sim \Lambda$: the nonlocal form factor modifies the effective potential, potentially suppressing high-energy emission.*
- *For astrophysical BH ($M \gg M_{\mathrm{Pl}}$): $T_H \sim 10^{-8}\,\mathrm{K}\cdot(M_\odot/M) \ll \Lambda$, so all emitted quanta satisfy $\omega \ll \Lambda$ and SCT corrections are completely negligible.*
- *The modified evaporation endpoint (where $T_H \sim \Lambda$ at $M \sim M_{\mathrm{Pl}}^2/\Lambda$) is where SCT corrections become significant. This connects to MR-9 (singularity resolution and information paradox).*

8 Entanglement Entropy Approach

[Bombelli et al. (1986), Phys. Rev. D **34**, 373; Srednicki (1993), Phys. Rev. Lett. **71**, 666.]

An alternative route to the BH entropy uses the entanglement entropy of quantum fields across the horizon:

$$S_{\mathrm{ent}} = -\operatorname{Tr}(\rho_{\mathrm{in}}\ln\rho_{\mathrm{in}}), \qquad \rho_{\mathrm{in}} = \operatorname{Tr}_{\mathrm{out}}\,|\Omega\rangle\langle\Omega|, \tag{36}$$

where $|\Omega\rangle$ is the vacuum state and the trace is over degrees of freedom outside the horizon.

Proposition 8.1 (Entanglement entropy and area law)**.** *[Bombelli et al. (1986); Srednicki (1993).] For a free scalar field on a flat background with a spherical entangling surface:*

$$S_{\mathrm{ent}} = c_1\,\frac{A}{\epsilon^2} + c_2\,\ln\!\left(\frac{A}{\epsilon^2}\right) + \mathcal{O}(1), \tag{37}$$

where ϵ is the UV cutoff. The area-law term is UV-divergent; the logarithmic term has a universal (cutoff-independent) coefficient.

Connection to SCT: In the spectral action framework, the UV cutoff is naturally provided by the spectral scale Λ. The entanglement entropy of the Dirac operator across the horizon should reproduce the Wald entropy in the semiclassical limit.

[GAP] The explicit computation of S_{ent} for the SCT spectral triple restricted to a causal diamond requires the near-horizon spectral geometry, which is an output of MT-1 (not an input). This is the "spectral action on horizon" approach (b) in the roadmap.

9 Benchmarks and Cross-Checks

Check	Expected	Source	Status
$S_{\text{EH}} = A/(4G)$	Bekenstein–Hawking	Wald (1993)	PASS
$\delta S_{C^2} = \alpha_C/\pi$	$13/(120\pi) \approx 0.035$	Jacobson–Myers (1993)	To verify
$\delta S_{R^2}\big\vert_{\text{Schw}} = 0$	$R = 0$	Direct	PASS
$c_{\log} = 37/24$	≈ 1.542	Sen (2012)	To verify
First law holds with S_{Wald}	Iyer–Wald (1994)		To verify
Zeroth law: $\kappa = \text{const}$	Standard GR proof	BCH (1973)	Conditional
Greybody $\to$ standard at $\omega \ll \Lambda$	Regge–Wheeler		To verify

10 Gaps and Open Questions

G1: Nonlocal Wald formula. The full Wald formula for a nonlocal Lagrangian $\int F(\Box)\, C^2$ requires integration by parts of the form factor away from the local limit. For astrophysical BH the local limit suffices, but for micro-BH this is essential. [GAP]

G2: Weyl binormal contraction. The contraction $C_{\mu\nu\rho\sigma}\, \hat{\epsilon}^{\mu\nu} \hat{\epsilon}^{\rho\sigma}$ on the Schwarzschild bifurcation surface must be computed explicitly in a regular coordinate system (Kruskal) and verified against the orthonormal-frame result (9). [needs check]

G3: Second law and ghost sector. Wall's proof requires well-posedness and linearized stability. The SCT ghost poles (MR-2) may violate these conditions. Status: CONDITIONAL on ghost resolution. [GAP]

G4: Entanglement entropy. The spectral triple approach to entanglement entropy across the horizon has not been computed explicitly for SCT. This is the second route (b) in the roadmap. [GAP]

G5: Kerr generalization. The Kerr BH has nonzero frame-dragging, and the Weyl binormal contraction differs from Schwarzschild. The R^2 sector still vanishes (Kerr is vacuum), but the topological correction α_C/π may receive angular momentum corrections. [needs check]

G6: Sen formula graviton contribution. The -424 in Sen's formula includes subtleties with the graviton zero modes on the Euclidean instanton. In SCT, the graviton propagator is modified (the Π_{TT} pole shifts), which may alter the zero-mode counting. [GAP]

G7: Modified evaporation endpoint. The late-stage evaporation (where $T_H \sim \Lambda$) requires the full nonlocal Regge–Wheeler equation (35) and connects to MR-9. [GAP]

G8: Information paradox. The spectral action's entire-function property may provide a natural resolution of the information paradox through modified near-horizon physics. This is deferred to MR-9. [GAP]

11 Master Reference Table

Ref	Citation	Key result	Use in MT-1
R1	Wald (1993), Phys. Rev. D **48**, R3427	BH entropy = Noether charge	Wald formula (10)
R2	Iyer–Wald (1994), Phys. Rev. D **50**, 846	First law for diff-invariant theories	First law (28)
R3	Jacobson–Myers (1993), PRL **70**, 3684	Entropy of Lovelock gravity	JM formula (17)
R4	Jacobson–Kang–Myers (1994), gr-qc/9312023	C^2 and R^2 Wald entropy	δS_{C^2}, δS_{R^2}
R5	Sen (2012), arXiv:1205.0971	$c_{\log}$ for non-extremal BH	Eq. (23)
R6	Wall (2015), arXiv:1504.08040	Second law for HD gravity	Section 6
R7	Bardeen–Carter–Hawking (1973)	Four laws of BH mechanics	Zeroth, first, third laws
R8	Hawking (1975), CMP **43**, 199	Hawking radiation	Section 7
R9	Regge–Wheeler (1957)	Perturbation equation	Greybody factors
R10	Buoninfante–Mazumdar (2020), 1903.01542	Nonlocal gravity + BH thermo	Local limit on horizon
R11	Conroy–Edholm–Mazumdar (2017), 1509.01636	Nonlocal Wald entropy	Nonlocal corrections
R12	Christensen–Duff (1978), NPB **154**, 301	Conformal anomaly coefficients	Relation to $c_{\log}$
R13	Stelle (1977), PRD **16**, 953	Higher-derivative gravity	Local limit benchmark
R14	Mann–Solodukhin (1998), hep-th/9709064	Conical singularity method	Alternative $c_{\log}$ derivation
R15	Bombelli et al. (1986), PRD **34**, 373	Entanglement entropy = area law	Entanglement approach
R16	Padmanabhan (2010), *Gravitation*	Weyl tensor on horizons	Prop. 2.2
R17	Kaul–Majumdar (2000), PRL **84**, 5255	LQG $c_{\log} = -3/2$	Comparison table
R18	Connes–Chamseddine (2010), 1004.0464	Spectral action and gravity	Spectral framework

12 Notation Summary

Symbol	Definition	Value/Source
α_C	Weyl^2 coefficient	13/120 (NT-1b Phase 3)
$\alpha_R(\xi)$	R^2 coefficient	$2(\xi - 1/6)^2$
$c_{\log}$	Logarithmic entropy correction	37/24 (Sen + SM, microcanonical)
κ	Surface gravity	$1/(4GM)$ (Schwarzschild)
T_H	Hawking temperature	$1/(8\pi GM)$
r_H	Horizon radius	$2GM$
A	Horizon area	$16\pi G^2 M^2$
$\hat{\epsilon}_{\mu\nu}$	Bifurcation surface binormal	$\hat{\epsilon}_{ab}\hat{\epsilon}^{ab} = -2$
$\chi(\mathcal{H})$	Euler characteristic	2 for S^2
N_s, N_f, N_v	SM field counts	$4, 45, 12$

13 Summary: The Full BH Entropy Formula in SCT

Collecting all contributions, the black hole entropy in SCT for a Schwarzschild BH is:

MT-1 Main Formula (to be verified)

$$S = \underbrace{\frac{A}{4G}}_{\text{Bekenstein–Hawking}} + \underbrace{\frac{13}{120\pi}}_{\text{Wald } (C^2\text{, topological})} + \underbrace{\frac{37}{24}\ln\left(\frac{A}{\ell_P^2}\right)}_{\text{one-loop (Sen)}} + \mathcal{O}(1). \tag{38}$$

The three terms have distinct physical origins:

1. $A/(4G)$: classical area law from the Einstein–Hilbert sector of the spectral action (Wald entropy).
2. $13/(120\pi)$: tree-level correction from the Weyl-squared sector; topological (depends on $\chi(\mathcal{H})$, not on A); parameter-free via $\alpha_C = 13/120$.
3. $(37/24)\ln(A/\ell_P^2)$: one-loop quantum correction from all SM fields + graviton; parameter-free via SM field content ($n_S = 4$, $n_F = 22.5$ Dirac, $n_V = 12$).

For an astrophysical BH ($M = 10\,M_\odot$): $S_{\rm BH} \approx 1.05 \times 10^{79}$, $\delta S_{\rm Wald} \approx 0.035$, $\delta S_{\log} \approx 283$. The hierarchy is $S_{\rm BH} \gg \delta S_{\log} \gg \delta S_{\rm Wald}$.

End of MT-1 Literature Extraction. All formulas extracted from cited sources; no original derivations.